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## SYSTEMS FOR POWER MODULE ASSEMBLY FOR INVERTER

### Abstract


Disclosed solutions relate to electrical inverters. In an example, a system including an inverter to convert DC power from a voltage source to AC power to drive a winding of a motor is disclosed. The inverter includes a first power module, which includes a first inverter to be connected to a first end of the winding and to be connected to the voltage source. The first inverter is configured to output first AC power to the first end of the winding. The first power module includes a second inverter to be connected to a second end of the winding and to be connected to the voltage source. The second inverter is configured to output second AC power to the second end of the winding. first power module includes a changeover switch to selectively connect and disconnect the second end of the winding to and from a neutral connection.



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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) [0001] This application claims the benefit of priority to U.S. Provisional Patent Application No. 63/562,673, filed Mar. 7, 2024, the entirety of which is incorporated by reference herein.

TECHNICAL FIELD
[0002] Various embodiments of the present disclosure relate generally to electrical inverters, and more particularly, but without limitation, to electrical inverters used in high voltage circuits such as electric vehicle applications.
BACKGROUND
[0003] Electrical inverters are widely used in high voltage applications, such as in electric vehicles. For example, electric vehicles typically employ multi-phase motors, which may use various winding configurations, which can be used to manipulate the torque and power output curves of a given motor by driving the motor in various configurations. Some inverter circuits for high voltage applications require additional high voltage cabling, complicated control, and generate excessive amounts of heat.
[0004] The present disclosure is directed to overcoming one or more of these challenges.
SUMMARY OF THE DISCLOSURE
[0005] In some aspects, the techniques described herein relate to a system including: an inverter to convert DC power from a voltage source to AC power to drive a motor, wherein the inverter includes: a first power module including: one or more first phase switches; one or more second phase switches; one or more positive DC power tabs connected to a positive DC rail to provide positive DC power to the one or more first phase switches and the one or more second phase switches; one or more negative DC power tabs connected to a negative DC rail to provide negative DC power to the one or more first phase switches and the one or more second phase switches; one or more first phase AC power tabs to receive AC power from the one or more first phase switches; and one or more second phase AC power tabs to receive AC power from the one or more second phase switches.
[0006] In some aspects, the techniques described herein relate to a system, wherein the first power module further includes: a flex circuit including an insulating material and an electrically conductive material.
[0007] In some aspects, the techniques described herein relate to a system, wherein the first power module further includes: a thermistor connected to the flex circuit.
[0008] In some aspects, the techniques described herein relate to a system, wherein the first power module further includes: a changeover switch to change a connection of the one or more first phase switches and the one or more second phase switches with a first winding of the motor.
[0009] In some aspects, the techniques described herein relate to a system, wherein the first power module further includes: a first substrate having an outer surface and an inner surface; a second substrate having an outer surface and an inner surface; and an electrically conductive spacer coupled to the inner surface of the first substrate and to the inner surface of the second substrate, wherein the one or more first phase switches and the one or more second phase switches are coupled to the inner surface of the first substrate and to the inner surface of the second substrate.
[0010] In some aspects, the techniques described herein relate to a system, wherein: the first power module is configured to drive a first AC phase of the motor, and the inverter further includes: a

second power module to drive a second AC phase of the motor; and a third power module to drive a third AC phase of the motor.

[0011] In some aspects, the techniques described herein relate to a system, wherein the inverter further includes: a capacitor.

[0012] In some aspects, the techniques described herein relate to a system, wherein the inverter further includes: a heat sink.

[0013] In some aspects, the techniques described herein relate to a system, further including: the voltage source to supply the DC power to the inverter; and the motor to receive the AC power from the inverter to drive the motor, wherein the system is provided as a vehicle including the inverter, the voltage source, and the motor.

[0014] In some aspects, the techniques described herein relate to a system including a power module for an inverter, the power module including: one or more first phase switches; one or more second phase switches; one or more positive DC power tabs connected to a positive DC rail to provide positive DC power to the one or more first phase switches and the one or more second phase switches; one or more negative DC power tabs connected to a negative DC rail to provide negative DC power to the one or more first phase switches and the one or more second phase switches; one or more first phase AC power tabs to receive AC power from the one or more first phase switches; and one or more second phase AC power tabs to receive AC power from the one or more second phase switches.

[0015] In some aspects, the techniques described herein relate to a system, wherein the power module further includes: a flex circuit including an insulating material and an electrically conductive material; and a thermistor connected to the flex circuit.

[0016] In some aspects, the techniques described herein relate to a system, wherein the power module further includes: a changeover switch to change a connection of the one or more first phase switches and the one or more second phase switches with a winding of a motor.

[0017] In some aspects, the techniques described herein relate to a system, wherein the power module further includes: one or more neutral power tabs.

[0018] In some aspects, the techniques described herein relate to a system, wherein one or more of the one or more first phase switches or the one or more second phase switches includes silicon carbide.

[0019] In some aspects, the techniques described herein relate to a system, wherein the power module further includes: one or more signal pins.

[0020] In some aspects, the techniques described herein relate to a system, wherein the power module further includes: a first substrate having an outer surface and an inner surface; a second substrate having an outer surface and an inner surface; and an electrically conductive spacer coupled to the inner surface of the first substrate and to the inner surface of the second substrate, wherein the one or more first phase switches and the one or more second phase switches are coupled to the inner surface of the first substrate and to the inner surface of the second substrate.

[0021] In some aspects, the techniques described herein relate to a system, wherein: the one or more first phase switches include a first upper phase switch and a first lower phase switch, the one or more second phase switches include a second upper phase switch and a second lower phase switch, the one or more positive DC power tabs include a first positive DC power tab for the first upper phase switch and a second positive DC power tab for the second upper phase switch, and the one or more negative DC power tabs include a first negative DC power tab for the first lower phase switch and a second negative DC power tab for the second lower phase switch.

[0022] In some aspects, the techniques described herein relate to a system including: an inverter to convert DC power from a voltage source to AC power to drive a motor, wherein the inverter includes: a first power module to drive a first AC phase of the motor; a second power module to drive a second AC phase of the motor; and a third power module to drive a third AC phase of the motor, wherein each of the first power module, the second power module, and the third power module includes: one or more first phase switches; one or more second phase switches; one or more positive DC power tabs connected to a positive DC rail to provide positive DC power to the one or more first phase switches and the one or more second phase switches; one or more negative DC power tabs connected to a negative DC rail to provide negative DC power to the one or more first phase switches and the one or more second phase switches; one or more first phase AC power tabs to receive AC power from the one or more first phase switches; and one or more second phase AC power tabs to receive AC power from the one or more second phase switches.

[0023] In some aspects, the techniques described herein relate to a system, wherein each of the first power module, the second power module, and the third power module further includes: one or more neutral power tabs.

[0024] In some aspects, the techniques described herein relate to a system, wherein the inverter further includes: a heat sink provided on a first side of the first power module, the second power module, and the third power module; and one or more capacitors provided on a second side of the first power module, the second power module, and the third power module, the second side opposite from the first side.

[0025] In some aspects, the techniques described herein relate to a system including: an inverter to convert DC power from a voltage source to AC power to drive a winding of a motor, wherein the inverter includes: a first power module including: a first inverter to be connected to a first end of the winding and to be connected to the voltage source, wherein the first inverter is configured to output first AC power to the first end of the winding; a second inverter to be connected to a second end of the winding and to be connected to the voltage source, wherein the second inverter is configured to output second AC power to the second end of the winding; and a changeover switch to selectively connect and disconnect the second end of the winding to and from a neutral connection.

[0026] In some aspects, the techniques described herein relate to a system, wherein the changeover switch is further configured to change a configuration of the winding from Delta to Wye or from Wye to Delta.

[0027] In some aspects, the techniques described herein relate to a system, wherein the first inverter is a first half bridge inverter and the second inverter is a second half bridge inverter.

[0028] In some aspects, the techniques described herein relate to a system, wherein: the first inverter includes one or more first upper phase switches and one or more first lower phase switches, and the second inverter includes one or more second upper phase switches and one or more second lower phase switches.

[0029] In some aspects, the techniques described herein relate to a system, wherein: the first inverter includes three upper phase switches and three lower phase switches to drive a first winding of the motor, and the second inverter includes three upper phase switches and three lower phase switches to drive the first winding of the motor.

[0030] In some aspects, the techniques described herein relate to a system, wherein the inverter further includes one or more controllers to control an operation of the first inverter, the second inverter, and the changeover switch.

[0031] In some aspects, the techniques described herein relate to a system, wherein: the first power module is configured to drive a first winding of the motor, and the inverter further includes: a second power module to drive a second winding of the motor; and a third power module to drive a third winding of the motor.

[0032] In some aspects, the techniques described herein relate to a system, wherein the inverter further includes: a capacitor.

[0033] In some aspects, the techniques described herein relate to a system, further including: the voltage source configured to supply the DC power to the inverter; and the motor configured to receive the AC power from the inverter to drive the motor, wherein the system is provided as a vehicle including the inverter, the voltage source, and the motor.

[0034] In some aspects, the techniques described herein relate to a system including a power module for an inverter, the power module including: a first inverter to be connected to a first end of a winding of a motor and to be connected to a voltage source, wherein the first inverter is configured to output first AC power to the first end of the winding; a second inverter to be connected to a second end of the winding and to be connected to the voltage source, wherein the second inverter is configured to output second AC power to the second end of the winding; and a changeover switch to selectively connect and disconnect the second end of the winding to and from a neutral connection.
[0035] In some aspects, the techniques described herein relate to a system, wherein the changeover switch is further configured to change a configuration of the winding from Delta to Wye or from Wye to Delta.
[0036] In some aspects, the techniques described herein relate to a system, wherein the first inverter is a first half bridge inverter and the second inverter is a second half bridge inverter.
[0037] In some aspects, the techniques described herein relate to a system, wherein: the first inverter includes three upper phase switches and three lower phase switches to drive the winding of the motor, and the second inverter includes three upper phase switches and three lower phase switches to drive the winding of the motor.
[0038] In some aspects, the techniques described herein relate to a system, wherein the changeover switch includes four switches.
[0039] In some aspects, the techniques described herein relate to a system, wherein the second inverter is configured to be disabled below a power threshold to reduce switching loss of the power module.
[0040] In some aspects, the techniques described herein relate to a system, wherein the changeover switch is configured to provide a current return path from the winding of the motor to the neutral connection when the second inverter is disabled.
[0041] In some aspects, the techniques described herein relate to a system, wherein the changeover switch is configured to provide a connection to the winding of the motor for boost voltage during charging of the voltage source.
[0042] In some aspects, the techniques described herein relate to a system including: an inverter to convert DC power from a voltage source to AC power to drive a motor, wherein the inverter includes: a first power module to drive a first winding of the motor; a second power module to drive a second winding of the motor; and a third power module to drive a third winding of the motor, wherein each of the first power module, the second power module, and the third power module includes: a first inverter to be connected to a first end of a respective winding of the motor and to be connected to a voltage source, wherein the first inverter is configured to output first AC power to the first end of the respective winding; a second inverter to be connected to a second end of the respective winding and to be connected to the voltage source, wherein the second inverter is configured to output second AC power to the second end of the respective winding; and a changeover switch to selectively connect and disconnect the second end of the respective winding to and from a neutral connection.
[0043] In some aspects, the techniques described herein relate to a system, wherein: each first inverter is a first half bridge inverter including three upper phase switches and three lower phase switches, and each second inverter is a second half bridge inverter including three upper phase switches and three lower phase switches.
[0044] In some aspects, the techniques described herein relate to a system, wherein the inverter is configured to operate in a first configuration with each second inverter is disabled, and a second configuration with each first inverter and each second inverter enabled.
[0045] In some aspects, the techniques described herein relate to a system including: an inverter to convert DC power from a voltage source to AC power to drive a motor, wherein the inverter includes: a capacitor assembly; and a first power module including: a first inverter to output first

AC power to a first end of a first winding of the motor; a second inverter to output second AC power to a second end of the first winding; and a changeover switch to selectively connect and disconnect the second end of the first winding to and from a neutral connection.
[0046] In some aspects, the techniques described herein relate to a system, wherein the capacitor assembly includes: one or more busbars; and one or more capacitors.
[0047] In some aspects, the techniques described herein relate to a system, wherein the one or more capacitors include: a first capacitor for the first inverter; and a second capacitor for the second inverter.
[0048] In some aspects, the techniques described herein relate to a system, wherein the one or more busbars include: a positive busbar connected to the one or more capacitors and the first power module; and a negative busbar connected to the one or more capacitors and the first power module.
[0049] In some aspects, the techniques described herein relate to a system, wherein the one or more busbars further include: a neutral busbar connected to the neutral connection of the first power module.
[0050] In some aspects, the techniques described herein relate to a system, wherein: the first power module is configured to drive the first winding of the motor, and the inverter further includes: a second power module to drive a second winding of the motor; and a third power module to drive a third winding of the motor.
[0051] In some aspects, the techniques described herein relate to a system, wherein the one or more capacitors include: a first capacitor for the first inverter of the first power module; a second capacitor for the second inverter of the first power module; a third capacitor for a first inverter of the second power module; a fourth capacitor for a second inverter of the second power module; a fifth capacitor for a first inverter of the third power module; and a sixth capacitor for a second inverter of the third power module.
[0052] In some aspects, the techniques described herein relate to a system, wherein: the positive busbar is further connected to the second power module and the third power module; and the negative busbar is further connected to the second power module and the third power module.
[0053] In some aspects, the techniques described herein relate to a system, further including: the voltage source configured to supply the DC power to the inverter; and the motor configured to receive the AC power from the inverter to drive the motor, wherein the system is provided as a vehicle including the inverter, the voltage source, and the motor.
[0054] In some aspects, the techniques described herein relate to a system including: an inverter to convert DC power from a voltage source to AC power to drive a motor, wherein the inverter includes: a capacitor assembly; and a power module including: a first inverter to output first AC power to a first end of a first winding of the motor; and a second inverter to output second AC power to a second end of the first winding.
[0055] In some aspects, the techniques described herein relate to a system, wherein the capacitor assembly includes: one or more busbars; and one or more capacitors.
[0056] In some aspects, the techniques described herein relate to a system, wherein the one or more capacitors include: a first capacitor for the first inverter; and a second capacitor for the second inverter.
[0057] In some aspects, the techniques described herein relate to a system, wherein the one or more busbars include: a positive busbar connected to the one or more capacitors and a positive connection of the power module; and a negative busbar connected to the one or more capacitors and a negative connection of the power module.
[0058] In some aspects, the techniques described herein relate to a system, wherein the one or more busbars further include: a neutral busbar connected to a neutral connection of the power module.
[0059] In some aspects, the techniques described herein relate to a system, wherein the power module includes: one or more positive power tabs connected to positive busbar; one or more negative power tabs connected to negative busbar; and one or more neutral tabs connected to

neutral busbar.
[0060] In some aspects, the techniques described herein relate to a system, wherein each of the positive busbar, the negative busbar, and the neutral busbar includes a respective connection tab on a same side of the capacitor assembly.
[0061] In some aspects, the techniques described herein relate to a system, wherein the power module further includes: a changeover switch to selectively connect and disconnect the second end of the first winding to and from a neutral connection.
[0062] In some aspects, the techniques described herein relate to a system including: an inverter to convert DC power from a voltage source to AC power to drive a motor, wherein the inverter includes: a capacitor assembly including one or more busbars and one or more capacitors; a first power module to drive a first winding of the motor; a second power module to drive a second winding of the motor; and a third power module to drive a third winding of the motor, wherein each of the first power module, the second power module, and the third power module includes: a first inverter to output first AC power to a first end of the respective winding; a second inverter to output second AC power to a second end of the respective winding; and a changeover switch to selectively connect and disconnect the second end of the respective winding to and from a neutral connection.
[0063] In some aspects, the techniques described herein relate to a system, wherein each of the first power module, the second power module, and the third power module further includes: one or more power tabs connected to the one or more capacitors via the one or more busbars.
[0064] In some aspects, the techniques described herein relate to a system, wherein the inverter further includes: a heat sink provided on a first side of the first power module, the second power module, and the third power module, wherein the capacitor assembly is provided on a second side of the first power module, the second power module, and the third power module, the second side opposite from the first side.
[0065] Additional objects and advantages of the disclosed embodiments will be set forth in part in the description that follows, and in part will be apparent from the description, or may be learned by practice of the disclosed embodiments. The objects and advantages of the disclosed embodiments will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims.
[0066] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosed embodiments, as claimed.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0067] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosed embodiments.
[0068] FIG. **1** depicts an exemplary system infrastructure for a power system for an electric vehicle, according to one or more embodiments.
[0069] FIG. **2** depicts an exemplary system infrastructure for a vehicle including an inverter, according to one or more embodiments.
[0070] FIG. **3** depicts an implementation of a computer system that may execute techniques presented herein, according to one or more embodiments.
[0071] FIG. **4** depicts an exemplary electrical schematic of a switchable inverter circuit, according to one or more embodiments.
[0072] FIG. **5** depicts a graph illustrating examples of torque and power output for various

configurations of motor windings, according to one or more embodiments.

[0073] FIG. **6** depicts an exemplary electrical schematic of a switchable inverter circuit configured in a Wye winding configuration, according to one or more embodiments.

[0074] FIG. **7** depicts an exemplary electrical schematic of a switchable inverter circuit configured in a Delta winding configuration, according to one or more embodiments.

[0075] FIG. **8** depicts an exemplary architecture for a power module, according to one or more embodiments.

[0076] FIG. **9** depicts an exemplary layout for a three-phase inverter package, according to one or more embodiments.

[0077] FIG. **10** depicts an exemplary physical construction of a power module, according to one or more embodiments.

[0078] FIG. **11** depicts an exemplary architecture for a power module, according to one or more embodiments.

[0079] FIG. **12** depicts an exemplary architecture for a power module, according to one or more embodiments.

[0080] FIG. **13** depicts an electrical schematic of an electrical inverter, according to one or more embodiments.

[0081] FIG. **14** depicts an exemplary layout for a three-phase inverter package, according to one or more embodiments.

[0082] FIG. **15** depicts an electrical schematic of an electrical inverter, according to one or more embodiments.

[0083] FIG. **16** depicts an electrical schematic of an electrical inverter, according to one or more embodiments.

[0084] FIG. **17** depicts an exemplary three-phase inverter package of an inverter circuit, according to one or more embodiments.

[0085] FIG. **18** depicts an exemplary architecture for a power module, according to one or more embodiments.

[0086] FIG. **19** depicts an exemplary architecture for a power module, according to one or more embodiments.

[0087] FIG. **20** depicts an assembly of a power module with devices on a single substrate, according to one or more embodiments.

[0088] FIG. **21** depicts an assembly of a power module with devices on a single substrate, according to one or more embodiments.

[0089] FIG. **22** depicts an assembly of a power module with devices on a single substrate, according to one or more embodiments.

[0090] FIG. **23** depicts an assembly of a power module with devices on a source substrate and a drain substrate, according to one or more embodiments.

[0091] FIG. **24** depicts an exemplary architecture for a power module including a thermistor, according to one or more embodiments.

## DETAILED DESCRIPTION OF EMBODIMENTS

[0092] Both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the features, as claimed. As used herein, the terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus. In this disclosure, unless stated otherwise, relative terms, such as, for example, “about,” “substantially,” and “approximately” are used to indicate a possible variation of ±10% in the stated value. In this disclosure, unless stated otherwise, any numeric value may include a possible variation of ±10% in the stated value.

[0093] The terminology used below may be interpreted in its broadest reasonable manner, even though it is being used in conjunction with a detailed description of certain specific examples of the present disclosure. Indeed, certain terms may even be emphasized below; however, any terminology intended to be interpreted in any restricted manner will be overtly and specifically defined as such in this Detailed Description section.

[0094] Various embodiments of the present disclosure relate generally to inverters, and more particularly, but without limitation, to inverter circuits used in high voltage circuits such as motors, or e-machines, in electric vehicles, and packages for inverter circuits.

[0095] As discussed above, electrical inverters are widely used in high voltage applications, such as for driving motors of electric vehicles. Motors are wound with electrical windings which may be connected with each other and with the power source(s) in various winding configurations. Examples include Wye ("Y"), Open Ended Winding (OEW), and Delta configurations. Switching winding configuration and/or applying one or both inverters of an open ended winding configuration may provide technical advantages such as an ability to adjust motor torque or power. But such switching circuits may pose disadvantages such as increased heat or length of high voltage cabling.

[0096] One or more embodiments may address these concerns. For example, disclosed systems include switchable inverter circuits that facilitate reconfiguration of inverters to and from different winding configurations, integrated power modules that include such switchable inverters, and integrated packages that include multiple power modules with thermal management. While the examples discussed herein describe electrical inverters for use in driving electric motors and/or in battery charging circuits, the disclosure is not limited thereto. Rather, the disclosed inverters may be used in any circuit.

[0097] Open coil motors may achieve similar power outputs using half the current required for some delta or Y connected motors. This is due to the ability to apply the full bus voltage across the winding rather than just half the bus voltage. In other words, one half of the sine wave is above the neutral point, and one half is below the neutral point. This may be done having the two inverters operating 180 degrees out of phase. However, the penalty for doing this is increased high voltage cabling, more complicated control, and a coordination problem during fault conditions. To address the coordination problem, both inverters may be installed within a single housing. Housing the inverters together allows more efficient fault safety logic and more tightly coupled control. However, increasing the number of switches also increases the required volume, and complicates the cooling and bus routing. Each of the six phases would have relative creepage and clearance requirements.

[0098] One or more embodiments may provide a highly integrated power switch and/or module topology that enables simplification of the internal dual inverter assembly, as a mitigation for increased complexity produced, with an open coil motor. One or more embodiments may provide a system partitioning, and a packaging strategy enabled by flex on substrate technology. The partitioning portion may group both coil A half bridges, both coil B half bridges, and both coil C half bridges. Each group is then integrated into a single power switch module.

[0099] One or more embodiments may provide a module that includes two half bridges including an upper power switch with three parallel SiC MOSFETs, a lower power switch with three parallel SiC MOSFETs, and a change-over switch with two sets (e.g., four switches) of parallel SiC MOSFETs in a back-to-back configuration. The back-to-back configuration may be used to block reverse current during the various operation scenarios. The N terminal change-over switch may only conduct when fully enhanced, when in mode two or three, or completely off as in mode one. The N terminal change-over switch is not intended for fast switching having fewer parallel MOSFETs.

[0100] The number of SiC FETs required per upper or lower section of the half bridge and change over switch may be dependent upon the total current and/or power requirement of the system. One or more embodiments may provide three parallel SiC die per switch. One or more embodiments may provide an operation mode one, with the highest power output, used during vehicle launch and at high speeds. During this scenario, both inverters are operational. The duty cycle controlling the sinusoidal current for inverter one would be 180 degrees out of phase from that controlling current on inverter two. Therefore, under static conditions, the duty cycle of inverter one is one minus the duty cycle of inverter two. Efficiency (conduction losses) in mode one, may be improved over some inverter topologies in that the voltage being switched across a SiC die would still be 800V, but the current would be half that of some inverters for comparable power delivery. The switch configuration may be operated under an alternate mode during reduced load situations, with the intent to increase efficiency.

[0101] One or more embodiments may provide an operation mode two, with low power output (e.g., below a power threshold), such as a vehicle operated at highway speeds under steady state conditions. The second inverter would be disabled, and the change-over switch is activated to provide a current return path. When NA is tied to NB and NC, with the second inverter disabled, the N connection will float to a voltage of the neutral point of a Y connected inverter at half the bus voltage.

[0102] Transitioning from mode one to mode two may require the forcing voltage produced by inverter two to decrease, while increasing the voltage produce by inverter one and maintaining the same inverter setpoint. When inverter two approaches 50% duty cycle, the second inverter may be disengaged, and the change-over switch engaged, which may reduce unwanted motor torque. Transitioning from mode two to mode one would be the opposite situation.

[0103] One or more embodiments may provide an operation mode three. If the vehicle was not in an operating mode, and the N terminal was provided with a positive voltage with respect to HV−, the change-over switch may be activated and inverter one may be used to boost the battery voltage. To maintain zero torque on the motor, the inverter may be duty cycled from active short circuit lower (all lower switches on) to active short circuit upper (all upper switches on).

[0104] One or more embodiments may provide an inverter design with NA tied to B2, NB tied to C2, and NC tied to A2. If inverter one is operational, inverter two is disabled, and with the change-over switch activated as in mode two. The inverter may drive the motor as if configured as a delta wound machine.

[0105] One or more embodiments may provide a switch module having an over molded exterior with features for increasing the creepage distance between terminals. One or more embodiments may provide two substrates having a thermally conductive interface to both sides of the SiC die. One or more embodiments may provide a second metal layer embedded within a polyimide flex circuit laminated onto the source substrate providing gate routing. One or more embodiments may provide a flex circuit having via features connecting flex signal trace to the underlying plane.

[0106] One or more embodiments may provide substrates carrying 16 SiC die with the sources connected to the underlying plane while the gate interconnect is connected to the flex circuit on second metal. The drains are then connected to the opposing substrate plane. Interconnects may be provided with soldering or sintering. One or more embodiments may provide a substrate to substrate interconnect using electrically conductive spacers. One or more embodiments may provide thermistors for thermal management. One or more embodiments may provide a lead frame providing interconnect to the rest of the inverter. One or more embodiments may provide an integrated architecture in a thick copper (high current), low inductance, double side cooled design.

[0107] One or more embodiments may provide a power module including three power module switch assemblies, and a cooling rail. One or more embodiments may provide a switch assembly sandwiched between two halves of the cooling rail. One or more embodiments may provide a power flow into the power connections, then into the capacitor assembly with integrated busbars, across the busbars, then into the terminals of the power module, and exiting the power module across the busbar assembly, then to the load at the phase terminals.

[0108] One or more embodiments may mitigate an increased complexity of the inverter assembly is with an increased number of interconnects. One or more embodiments may provide a bulk capacitor assembly with busbars with an additional plate in order to route the N signal away from the power modules. One or more embodiments may provide a phase connection busbar assembly that accommodates cross over interconnects if inverter one and inverter two outputs are ordered by inverter.
[0109] One or more embodiments may provide an open coil Y connected motor being driven by two inverters. One or more embodiments may provide packaging for a two-inverter solution for an open coil motor design. One or more embodiments may provide a two inverter open coil design. One or more embodiments may provide a power switch configuration with a switch for turning off one or the other inverters to save switching loss at lower power levels. One or more embodiments may provide a power switch configuration that provides a current return path for the motor winding to a neutral path in the case that the motor is being driven by only one inverter. One or more embodiments may provide a switch to provide a path through the motor winding for boost voltage during charging.
[0110] One or more embodiments may provide reduced cooling rails and interconnects. One or more embodiments may provide reduced overall volume. One or more embodiments may provide power flows in from the external terminals on the right side into the capacitor. One or more embodiments may provide a bulk capacitor assembly that has busbars with an additional plate in order to route the N signal away from the power modules. One or more embodiments may provide a path where power flows into the power switches and is chopped into a sinusoidal voltage that is then applied to both sides of the coil loads. One or more embodiments may provide an integrated module that allows for decreased volume. One or more embodiments may provide lower loop inductance that may provide better switching characteristics. One or more embodiments may provide a highly integrated switch topology using flex on substrate. One or more embodiments may provide a bulk capacitor with an additional plate.
[0111] FIG. **1** depicts an exemplary system infrastructure for a power system for an electric vehicle, according to one or more embodiments. As explained herein, power system **100** may perform functions such as charging battery pack **140** or powering an electric vehicle via motor **190**.
[0112] As depicted in FIG. **1**, power system **100** may include or be electrically connectable to a charging connector **110**. The charging connector **110** may provide an electrical connection from an external power supply to the power system **100**, and may be a Type 1 or a Type 2 connector, for example. The charging connector **110** may transfer single phase, two-phase, or three phase power.
[0113] The power system **100** may include one or more of an inverter **120**, an HV DC-DC converter **130**, and a controller **300** receiving signals from input sensor **150**. Inverter **120** may convert DC, for example, provided by battery pack **140**, to AC in one or more phases for driving motor **190**. As explained herein, various inverter configurations are possible. For instance, power system **100** may include circuits as depicted herein. As discussed, these circuits may reconfigure a connection of the windings used to energize motor **190** as needs change. In some cases, power system **100** may include a Power Factor Correction (PFC) converter (not depicted). The PFC converter may be an AC-DC converter. HV DC-DC converter **130** may be a DC-DC converter.
[0114] Controller **300** may include one or more controllers such as processors. The power system **100** may include or be electrically connectable to a battery pack **140**. The power system **100** may be used in automotive vehicles as an onboard charger to transfer power from an external power source through charging connector **110** to battery pack **140** in a grid-to-battery operation, or to transfer power from battery pack **140** in a vehicle to grid configuration (a battery-to-grid operation). The power system **100** may be included in a system provided as an electric vehicle including a motor **190** configured to rotate based on power received from the battery pack **140**.
[0115] FIG. **2** depicts an exemplary system infrastructure for a vehicle including a battery charger, according to one or more embodiments. As discussed, power system **100** may include an inverter

and/or a battery charger. As depicted in FIG. **2**, electric vehicle **185** may include power system **100**, motor **190**, and battery pack **140**.

[0116] Power system **100** may include components to receive electrical power from an external source and output electrical power to charge battery pack **140** of electric vehicle **185**. Power system **100** may convert DC power from battery pack **140** in electric vehicle **185** to AC power, to drive motor **190** of the electric vehicle **185**, for example, but the embodiments are not limited thereto. For example, power system **100** may include components to receive electrical power from an external source and output electrical power to charge battery pack **140** without motor **190** connected to power system **100**. Power system **100** may convert DC power from battery pack **140** in electric vehicle **185** to AC power, to drive AC components other than motor **190** of the electric vehicle **185**. Power system **100** may be bidirectional, and may convert DC power to AC power, or convert AC power to DC power, such as during regenerative braking, for example. Power system **100** may be a three-phase inverter, a single-phase inverter, or a multi-phase inverter.

[0117] FIG. **3** depicts an implementation of a controller **300** that may execute techniques presented herein, according to one or more embodiments. For example, controller **300** may control one or more configurations of the circuits discussed herein.

[0118] Any suitable system infrastructure may be put into place to allow control of the battery charger. FIG. **3** and the following discussion provide a brief, general description of a suitable computing environment in which the present disclosure may be implemented. In one embodiment, any of the disclosed systems, methods, and/or graphical user interfaces may be executed by or implemented by a computing system consistent with or similar to that depicted in FIG. **3**. Although not required, embodiments of the present disclosure are described in the context of computer-executable instructions, such as routines executed by a data processing device, e.g., a server computer, wireless device, and/or personal computer. Those skilled in the relevant art will appreciate that embodiments of the present disclosure can be practiced with other communications, data processing, or computer system configurations, including: Internet appliances, hand-held devices (including personal digital assistants (“PDAs”)), wearable computers, all manner of cellular or mobile phones (including Voice over IP (“VoIP”) phones), dumb terminals, media players, gaming devices, virtual reality devices, multi-processor systems, microprocessor-based or programmable consumer electronics, set-top boxes, network PCs, mini-computers, mainframe computers, and the like. Indeed, the terms “computer,” “server,” and the like, are generally used interchangeably herein, and refer to any of the above devices and systems, as well as any data processor.

[0119] Embodiments of the present disclosure may be embodied in a special purpose computer and/or data processor that is specifically programmed, configured, and/or constructed to perform one or more of the computer-executable instructions explained in detail herein. While embodiments of the present disclosure, such as certain functions, are described as being performed exclusively on a single device, the present disclosure may also be practiced in distributed environments where functions or modules are shared among disparate processing devices, which are linked through a communications network, such as a Local Area Network (“LAN”), Wide Area Network (“WAN”), and/or the Internet. Similarly, techniques presented herein as involving multiple devices may be implemented in a single device. In a distributed computing environment, program modules may be located in both local and/or remote memory storage devices.

[0120] Embodiments of the present disclosure may be stored and/or distributed on non-transitory computer-readable media, including magnetically or optically readable computer discs, hard-wired or preprogrammed chips (e.g., EEPROM semiconductor chips), nanotechnology memory, biological memory, or other data storage media.

[0121] Alternatively, computer implemented instructions, data structures, screen displays, and other data under embodiments of the present disclosure may be distributed over the Internet and/or over other networks (including wireless networks), on a propagated signal on a propagation medium (e.g., an electromagnetic wave(s), a sound wave, etc.) over a period of time, and/or they may be provided on any analog or digital network (packet switched, circuit switched, or other scheme).

[0122] The controller **300** may include a set of instructions that may be executed to cause the controller **300** to perform any one or more of the methods or computer-based functions disclosed herein. The controller **300** may operate as a standalone device or may be connected, e.g., using a network, to other computer systems or peripheral devices.

[0123] In a networked deployment, the controller **300** may operate in the capacity of a server or as a client in a server-client user network environment, or as a peer computer system in a peer-to-peer (or distributed) network environment. The controller **300** may also be implemented as or incorporated into various devices, such as a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a mobile device, a palmtop computer, a laptop computer, a desktop computer, a communications device, a wireless telephone, a land-line telephone, a control system, a camera, a scanner, a facsimile machine, a printer, a pager, a personal trusted device, a web appliance, a network router, switch or bridge, or any other machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. In a particular implementation, the controller **300** may be implemented using electronic devices that provide voice, video, or data communication. Further, while the controller **300** is illustrated as a single system, the term "system" shall also be taken to include any collection of systems or sub-systems that individually or jointly execute a set, or multiple sets, of instructions to perform one or more computer functions.

[0124] As illustrated in FIG. **3**, the controller **300** may include a processor **302**, e.g., a central processing unit (CPU), a graphics processing unit (GPU), or both. The processor **302** may be a component in a variety of systems. For example, the processor **302** may be part of a standard computer. The processor **302** may be one or more general processors, digital signal processors, application specific integrated circuits, field programmable gate arrays, servers, networks, digital circuits, analog circuits, combinations thereof, or other now known or later developed devices for analyzing and processing data. The processor **302** may implement a software program, such as code generated manually (i.e., programmed).

[0125] The controller **300** may include a memory **304** that may communicate via a bus **308**. The memory **304** may be a main memory, a static memory, or a dynamic memory. The memory **304** may include, but is not limited to computer readable storage media such as various types of volatile and non-volatile storage media, including but not limited to random access memory, read-only memory, programmable read-only memory, electrically programmable read-only memory, electrically erasable read-only memory, flash memory, magnetic tape or disk, optical media and the like. In one implementation, the memory **304** includes a cache or random-access memory for the processor **302**. In alternative implementations, the memory **304** is separate from the processor **302**, such as a cache memory of a processor, the system memory, or other memory. The memory **304** may be an external storage device or database for storing data. Examples include a hard drive, compact disc ("CD"), digital video disc ("DVD"), memory card, memory stick, floppy disc, universal serial bus ("USB") memory device, or any other device operative to store data. The memory **304** is operable to store instructions executable by the processor **302**. The functions, acts or tasks illustrated in the figures or described herein may be performed by the processor **302** executing the instructions stored in the memory **304**. The functions, acts or tasks are independent of the particular type of instructions set, storage media, processor or processing strategy and may be performed by software, hardware, integrated circuits, firm-ware, micro-code and the like, operating alone or in combination. Likewise, processing strategies may include multiprocessing, multitasking, parallel processing and the like.

[0126] As depicted, the controller **300** may further include a display **310**, such as a liquid crystal display (LCD), an organic light emitting diode (OLED), a flat panel display, a solid-state display, a cathode ray tube (CRT), a projector, a printer or other now known or later developed display device for outputting determined information. The display **310** may act as an interface for the user to see the functioning of the processor **302**, or specifically as an interface with the software stored in the memory **304** or in the drive unit **306**.

[0127] Additionally or alternatively, the controller **300** may include an input device **312** configured to allow a user to interact with any of the components of controller **300**. The input device **312** may be a number pad, a keyboard, or a cursor control device, such as a mouse, or a joystick, touch screen display, remote control, or any other device operative to interact with the controller **300**.

[0128] The controller **300** may also or alternatively include drive unit **306** implemented as a disk or optical drive. The drive unit **306** may include a computer-readable medium **322** in which one or more sets of instructions **324**, e.g. software, may be embedded. Further, the instructions **324** may embody one or more of the methods or logic as described herein. The instructions **324** may reside completely or partially within the memory **304** and/or within the processor **302** during execution by the controller **300**. The memory **304** and the processor **302** also may include computer-readable media as discussed above.

[0129] In some systems, a computer-readable medium **322** includes instructions **324** or receives and executes instructions **324** responsive to a propagated signal so that a device connected to a network **370** may communicate voice, video, audio, images, or any other data over the network **370**. Further, the instructions **324** may be transmitted or received over the network **370** via a communication port or interface **320**, and/or using a bus **308**. The communication port or interface **320** may be a part of the processor **302** or may be a separate component. The communication port or interface **320** may be created in software or may be a physical connection in hardware. The communication port or interface **320** may be configured to connect with a network **370**, external media, the display **310**, or any other components in controller **300**, or combinations thereof. The connection with the network **370** may be a physical connection, such as a wired Ethernet connection or may be established wirelessly as discussed below. Likewise, the additional connections with other components of the controller **300** may be physical connections or may be established wirelessly. The network **370** may alternatively be directly connected to a bus **308**.

[0130] While the computer-readable medium **322** is shown to be a single medium, the term "computer-readable medium" may include a single medium or multiple media, such as a centralized or distributed database, and/or associated caches and servers that store one or more sets of instructions. The term "computer-readable medium" may also include any medium that is capable of storing, encoding, or carrying a set of instructions for execution by a processor or that cause a computer system to perform any one or more of the methods or operations disclosed herein. The computer-readable medium **322** may be non-transitory, and may be tangible.

[0131] The computer-readable medium **322** may include a solid-state memory such as a memory card or other package that houses one or more non-volatile read-only memories. The computer-readable medium **322** may be a random-access memory or other volatile re-writable memory. Additionally or alternatively, the computer-readable medium **322** may include a magneto-optical or optical medium, such as a disk or tapes or other storage device to capture carrier wave signals such as a signal communicated over a transmission medium. A digital file attachment to an e-mail or other self-contained information archive or set of archives may be considered a distribution medium that is a tangible storage medium. Accordingly, the disclosure is considered to include any one or more of a computer-readable medium or a distribution medium and other equivalents and successor media, in which data or instructions may be stored.

[0132] In an alternative implementation, dedicated hardware implementations, such as application specific integrated circuits, programmable logic arrays and other hardware devices, may be constructed to implement one or more of the methods described herein. Applications that may include the apparatus and systems of various implementations may broadly include a variety of electronic and computer systems. One or more implementations described herein may implement functions using two or more specific interconnected hardware modules or devices with related control and data signals that may be communicated between and through the modules, or as portions of an application-specific integrated circuit. Accordingly, the present system encompasses software, firmware, and hardware implementations.

[0133] The controller **300** may be connected to a network **370**. The network **370** may define one or more networks including wired or wireless networks. The wireless network may be a cellular telephone network, an 802.11, 802.16, 802.20, or WiMAX network. Further, such networks may include a public network, such as the Internet, a private network, such as an intranet, or combinations thereof, and may utilize a variety of networking protocols now available or later developed including, but not limited to TCP/IP based networking protocols. The network **370** may include wide area networks (WAN), such as the Internet, local area networks (LAN), campus area networks, metropolitan area networks, a direct connection such as through a Universal Serial Bus (USB) port, or any other networks that may allow for data communication. The network **370** may be configured to couple one computing device to another computing device to enable communication of data between the devices. The network **370** may generally be enabled to employ any form of machine-readable media for communicating information from one device to another. The network **370** may include communication methods by which information may travel between computing devices. The network **370** may be divided into sub-networks. The sub-networks may allow access to all of the other components connected thereto or the sub-networks may restrict access between the components. The network **370** may be regarded as a public or private network connection and may include, for example, a virtual private network or an encryption or other security mechanism employed over the public Internet, or the like.

[0134] In accordance with various implementations of the present disclosure, the methods described herein may be implemented by software programs executable by a computer system. Further, in an exemplary, non-limited implementation, implementations may include distributed processing, component/object distributed processing, and parallel processing. Alternatively, virtual computer system processing may be constructed to implement one or more of the methods or functionality as described herein.

[0135] Although the present specification describes components and functions that may be implemented in particular implementations with reference to particular standards and protocols, the disclosure is not limited to such standards and protocols. For example, standards for Internet and other packet switched network transmission (e.g., TCP/IP, UDP/IP, HTML, and HTTP) represent examples of the state of the art. Such standards are periodically superseded by faster or more efficient equivalents having essentially the same functions. Accordingly, replacement standards and protocols having the same or similar functions as those disclosed herein are considered equivalents thereof.

[0136] It will be understood that the steps of methods discussed are performed in one embodiment by an appropriate processor (or processors) of a processing (i.e., computer) system executing instructions (computer-readable code) stored in storage. It will also be understood that the disclosure is not limited to any particular implementation or programming technique and that the disclosure may be implemented using any appropriate techniques for implementing the functionality described herein. The disclosure is not limited to any particular programming language or operating system.

[0137] As discussed, certain aspects relate to improved inverters and power modules that may simplify electronics and improve thermal management associated with the switchable configuration of open coil windings and various winding configurations. In so doing, certain aspects may enable improved electric vehicle systems.

[0138] FIG. **4** depicts an exemplary electrical schematic of a switchable inverter circuit **400**, according to one or more embodiments. In the example depicted, switchable inverter circuit **400** (also referred to as a power switch module) uses changeover switch **430** to manage operation of first inverter **410**, second inverter **420**, and a configuration of motor windings **440**. Inverter circuit

**400** may be reconfigured, via changeover switch **430**, to provide power to motor windings **440** in either a Wye or a Delta winding configuration. Inverter circuit **400** may be reconfigured, also via changeover switch **430**, to provide power from one or both of first inverter **410** and second inverter **420**. Changeover switch **430** may selectively connect and disconnect the second end of the motor windings **440** to and from a neutral connection.

[0139] In a first use case, changeover switch **430** configures motor windings **440** to be in a Wye mode, such that the motor windings **440** receive power from both the first inverter **410** and second inverter **420**. In this mode, the voltages provided by first inverter **410** and second inverter **420** may be out-of-phase from each other. In this case, the A1 and B2 connection points are connected, the B1 and C2 connection points are connected, and the C1 and A2 connection points are connected.

[0140] In a second use case, changeover switch **430** configures motor windings **440** to be in a Delta mode, receiving power from only the first inverter **410**. In this case, the second inverter **420** is bypassed. Other use cases are possible. Different winding configurations may have different technical advantages. For example, the Wye configuration may be used at lower motor speeds, whereas the Delta configuration may be used at higher speeds. FIG. **5** illustrates some of these advantages.

[0141] FIG. **5** depicts a graph **500** illustrating examples of torque and power output for various configurations of motor windings, according to one or more embodiments. Graph **500** depicts both torque (left) and power (right) of both Wye and Delta winding configurations. As can be seen, the Wye configuration may provide greater torque at low rotational speeds than the Delta configuration. But at higher speed, the Delta configuration may provide increased power relative to the Wye configuration. Accordingly, disclosed systems may adjust the winding configuration as needed.

[0142] FIG. **6** depicts an exemplary electrical schematic of a switchable inverter circuit **600** configured in a Wye winding configuration, according to one or more embodiments. Inverter circuit **600** may include power switch module **610**, power switch module **612**, power switch module **614**, motor windings **670**, and power source **680**.

[0143] Each power switch module **610**, **612**, and **614** may include two inverters (phase switches) and a changeover switch. Examples of suitable inverters include half bridge inverters, but other inverter types are possible.

[0144] For example, power switch module **610** may include half bridge modules **620** and **625**. In turn, half bridge module **620** may include upper power switch **630** and lower power switch **640**. Similarly, half bridge module **625** may include upper power switch **635** and lower power switch **645**.

[0145] Power switch module **610** may include changeover switch **650**, which may adjust voltages from half bridge modules **620** and/or **625**. For instance, in the Wye configuration as depicted, power from only half bridge module **620** may be applied to motor windings **670** and power from half bridge module **625** is not applied to motor windings **670**.

[0146] As further discussed with respect to FIG. **7**, switchable inverter circuit **600** may be reconfigured to use a Delta configuration with both half-bridge modules **620** and **625** active.

[0147] FIG. **7** depicts an exemplary electrical schematic of a switchable inverter circuit **700** configured in a Delta winding configuration, according to one or more embodiments. Inverter circuit **700** may include power switch module **710**, power switch module **712**, power switch module **714**, motor windings **770**, and power source **780**. Relative to FIG. **6**, motor windings **770** are configured in a Delta configuration rather than in a Wye configuration (see motor windings **670**).

[0148] Traditionally, an implementation of a changeover switch may be complex to build and may require additional cooling, for instance, three separate cooling rails, one for each phase. By contrast, one or more embodiments may address these concerns by partitioning the system such that a single phase inverter and single changeover switch are provided in close proximity. For example, a given power switch module (for instance, **610**, **612**, **614**, **710**, **712**, and **714**) may be integrated into a single package that may include two half bridge inverters and a changeover switch in close proximity. This approach may shorten high voltage connections and reduce heat. One such example is discussed with respect to FIG. **8**.

[0149] FIG. **8** depicts an exemplary architecture for a power module **800**, according to one or more embodiments. Power module **800** represents an integrated package that may include two inverters (e.g., half-bridge inverters) and a changeover circuits, as discussed with respect to FIGS. **6** and **7**. A given power module **800** therefore represents one phase of a multi-phase power delivery system. As discussed further herein, for instance with respect to FIG. **9**, multiple power modules may be integrated into a single system in conjunction with a cooling system.

[0150] By placing inverters and a changeover switch for a given phase in a single package, certain advantages may be obtained such as lower heat and shorter high voltage paths. To realize these advantages, power module **800** may leverage various technologies such as “flex on substrate” (FoS). A substrate is a non-conductive material that provides mechanical support and electrical insulation for the components and conductive traces. FoS may allow a full use of the source thermal path by using openings in a dielectric to directly make a connection to the lower substrate and may employ thin layers of polyimide routed under a die to make gate connections without interference with source connections. Disclosed power modules may also leverage features overmolded exteriors, as well as various features such as electrically conductive spacers, vias, and surface mounted thermistors.

[0151] Power module **800** may further include exterior **801**, feature **802**, substrates **803** and **804**, metal layer **805**, vias **806**, substrates **807**, electrically conductive spacers **808**, thermistors **809**, tabs **810**, inverter **810**, inverter **820**, and changeover switch **840**.

[0152] Examples of inverter **820**, inverter **830**, and changeover switch **840** are discussed in FIGS. **6** and **7**. Inverter **820** and inverter **830** may be half-bridge inverters. These Inverters may be used to apply pulse width modulated voltage in controlling the current applied to inductive loads such as motors. As depicted, inverter **820** is positioned towards the left side of the power module **800**, inverter **830** to the middle, and changeover switch **840** towards the right of power module **800**. But other arrangements are possible.

[0153] Power module **800** may be formed of one or more substrates, metal layers, and so forth. In one example, a source and its related circuits may most naturally be on the opposing substrates for the two switches in a half bridge inverter. In a half bridge circuit, a source of an upper switch and a drain of the lower switch may share a common connection with the load, and the source and drain may be on opposite sides of a bare die that are used in power applications. For high voltage, dual side cooled applications, the source and the drain may be cooled at their interconnects, through an insulating substrate, with the drain being on one substrate and the source being on the other. Other layouts are possible.

[0154] Exterior **801** may be overmolded and may comprise a dielectric material. Feature **802**, shown as a carveout of exterior **801**, may serve to increase creepage distance between terminals or tabs. A creepage distance is a shortest distance between two conductive paths measured along the surface of a solid insulation.

[0155] Power module **800** may also include one or more metal layers **805** and/or substrates **807**. For instance, substrates **803** and **804** may have a thermally conductive interface to both sides of the SiC die. Substrate **803** may be a Si3N4 substrate with AMB copper providing 1st metal layer. Substrate **804**, may be a Si3N4 substrate with AMB copper providing 3.sup.rd metal layer. Metal layer **805** may be embedded within a polyimide flex circuit laminated onto the source substrate providing gate routing. In some cases, metal layer **805** may include multiple conductive regions electrically separated from each other.

[0156] Substrate **807** may be positioned to carry a Silicon Carbide (SiC) die. In an example, sources are connected to the underlying plane while the gate interconnect is to the flex circuit on second metal. The drains may be connected to the opposing substrate plane. Interconnect may be done with soldering or sintering. Substrate **807** may have silicon nitride (Si.sub.3N.sub.4) middle layers having thick metallization, e.g., direct bond copper (DBC) or active metal brazing (AMB) that may be employed on an outer surface and an inner surface of a ceramic middle layer. The SiC die may have a drain connection to the upper substrate and/or a source connection attached to the lower substrate.

[0157] In an example, power module **800** may include a Si3N4 substrate with AMB copper providing a metal layer (e.g., a first metal layer) and/or a Si3N4 substrate with AMB copper providing a metal layer (e.g., a third metal layer), and/or a flex on substrate layer providing a metal layer (e.g., a second metal layer) on the source side. Additional details are discussed further with respect to FIG. **24**.

[0158] Power module **800** may include one or more features such as vias **806** connecting flex signal trace to an underlying plane. In some cases, a via may connect to or from one or more of the conducting layers.

[0159] Electrically conductive spacers **808** may provide substrate-to-substrate interconnects. Spacers may help resolve assembly challenges including heat and current paths. The spacers may be formed of copper, for example, or another electrically and/or thermally conductive material. The spacers may provide a connection between a first substrate and a second substrate, thereby conducting current between substrates.

[0160] In an example, a spacer is approximately 2 mm by approximately 2 mm, and may have a thickness similar to that of a die, such as approximately 180 μm, for example. The spacers may be surface mount components and placed along with SiC MOSFETs using a same process. The spacers may replace wires and clips that some systems may use, and die interconnects may allow the gate, drain, and source to be connected to the same substrate.

[0161] The spacers may provide for reduced loop inductance in the device. Reduced loop inductance may, in turn, provide less ringing and less voltage overshoot. An application of the device may then speed up the switching time, reduce the power, and may offer more current throughput at a given price and performance point.

[0162] The spacers may be compatible with methods used to make connections to both sides of a die, such as by sinter or solder interconnect, for example. The electrically conductive spacers may thus allow current to flow between the two substrates.

[0163] In an aspect, power module **800** may include one or more surface-mount components such as, for example, a temperature sensor, thermistor, capacitor, resistor, or another integrated circuit (IC), such as a gate driver, etc. As depicted, thermistors **809** are provided for thermal management. For example, a temperature measurement from thermistors **809** may be provided to controller **300**. In turn, controller **300** may take one or more actions such as adjusting current, changing winding configuration, sending an alert, and so forth.

[0164] Examples of tabs **810**, or connections include various voltage, or power tabs. For example, tabs **810** include positive voltage (HV1+) **851**, negative voltage (HV1−) **852**, negative voltage (HV2−) **853**, positive voltage (HV2+) **854**, neutral (N) **855**, phase 1 output (PH1) **861**, and phase 2 output (PH2) **862**. In an example, DC power is provided to positive voltage (HV1+) **851**, negative voltage (HV1−) **852**, negative voltage (HV2−) **853**, and positive voltage (HV2+) **854**. In turn, the inverter **820** and inverter **830** output AC power to phase 1 output (PH1) **861**, and phase 2 output (PH2) **862** respectively. Tabs **810** may be formed of any conductive material such as copper.

[0165] Power module **800** may include a lead frame that provides interconnect to the rest of the inverter. The lead frame may provide improvements relative to existing technologies.

[0166] As discussed further herein, various cooling systems, active and/or passive may be used. For example, a cooling system may include an active cooling system, e.g., an active heat sink, and a passive cooling system, e.g., a passive heat sink having a thermo-conductive cooling structure. The thermo-conductive cooling structure may include a plate of thermally conductive materials, as described in more detail below. The passive heat sink (e.g., thermo-conductive cooling structure) may contact both the active heat sink and a power switch. The passive heat sink (e.g., thermo-conductive cooling structure) may be configured to dissipate heat away from the power switch and toward the active heat sink.

[0167] Additional examples of power modules are discussed further with respect to FIGS. **11**, **12**, **18**, and **19**. Other examples are possible.

[0168] FIG. **9** depicts an exemplary three-phase inverter package **900**, according to one or more embodiments. Package **900** includes a case **930** which facilitates connections with power modules **920**, **922**, and **924**, which in turn are actively cooled by cooling rail **970**. An example of power modules **920**, **922**, and **924** is power module **800** depicted in FIG. **8**. Power modules may also be referred to as switch assemblies.

[0169] Case **930** facilitates connection of the positive rail, negative rail, and neutral rail to each of power module **920**, **922**, and **924**. DC power is applied to the positive and negative rail, flows across busbars and into the terminals of the power module package **900**. Power exits the power module across the busbar assembly **960** via the phase terminals **905** to the electrical load (e.g., motor).

[0170] Case **930** may include positive voltage input **901**, negative voltage input **902**, and neutral output **903**. These connections are passed across one or more plates on case **930**. The plates connect to the power modules via various terminals **904**. For example, a first plate may conduct power from positive voltage input **901** to the power modules. A second plate may conduct power from negative voltage input **902** to the power modules. A third plate may conduct power from the power modules to neutral output **903**.

[0171] In turn, each of power modules **920**, **922**, and **924** converts DC to AC power and outputs AC power via one or more phase terminals **905**. Phase terminals **905** are assembled onto busbar assembly **960**.

[0172] The neutral output **903** is a generated output by the inverters. Access to the neutral rail may provide advantages in the case of a failure event. The power modules **920**, **922**, and **924** may be cooled via an active and/or passive cooling system. For example, fluid **980** may enter cooling rail **970**, cool the power modules, and be returned. Cooling rail **970** may include two halves (not depicted).

[0173] Case **930** may include one or more capacitors **910** (e.g., six capacitors). Each capacitor (or capacitor pair, for example) may correspond to one of power modules **920**, **922**, and **924**.

[0174] FIG. **10** depicts an exemplary physical construction **1000** of a power module, according to one or more embodiments. FIG. **10** depicts assembly of a power module including use of a substrate. Physical construction **1000** may include stages **1001**, **1002**, **1003**, **1004**, and **1005**, but other stages are possible.

[0175] Each stage **1000-1005** represents the addition of one or more components to the assembly. Physical construction **1000** may further include substrate **1010**, flex material **1011** (which may include a second metal layer), solder preforms **1012**, various components **1013**, additional solder preforms **1014**, and drain substrate **1015**. Examples of substrate include Si3N4 AMB SBST.

[0176] At stage **1001**, substrate **1010** is integrated with flex material **1011**. At stage **1002**, the integration created at stage **1001**, e.g., substrate **1010** and flex material **1011**, is integrated with solder preforms **1012**. At stage **1003**, the integration created at stage **1002** is integrated with components **1013**. At stage **1004**, the integration created at stage **1003** is integrated with solder preforms **1014**. At stage **1005**, the integration created at stage **1004** is integrated with drain substrate **1015**. In some cases, the substrate **1010** and drain substrate **1015** may be individually assembled, and then integrated to form a power module.

[0177] The assembly may include laminating polyimide film onto the source substrate. The polyimide, carrying the second metal, may replace a solder mask on the source substrate. The assembly may include using placed informs rather than a solder print operation, and may provide a

consistency in height necessary to reduce a variation in thickness. Placed informs may also allow introduction of a higher temperature solder. The power module may provide a significant increase in current capability relative to some systems.
[0178] FIG. **11** depicts an exemplary architecture for a power module **1100**, according to one or more embodiments. Power module **1100** may include one or more inverters and/or changeover switches. Relative to power module **800** as depicted in FIG. **8**, power module **1100** does not include a neutral tab
[0179] As depicted, power module **1100** may include various connections or tabs, such as positive voltage (HV1+) **1151**, negative voltage (HV1−) **1152**, negative voltage (HV2−) **1153**, positive voltage (HV2+) **1154**, phase 1 output (PH1) **1161**, and phase 2 output (PH2) **1162**.
[0180] FIG. **12** depicts an exemplary architecture for a power module **1200**, according to one or more embodiments. Power module **1200** may include one or more inverters and/or changeover switches. Relative to power module **800** as depicted in FIG. **8**, power module does not include a neutral tab and includes a CP tab.
[0181] As depicted, power module **1200** may include various connections or tabs, such as positive voltage (HV1+) **1251**, negative voltage (HV1−) **1252**, negative voltage (HV2−) **1253**, positive voltage (HV2+) **1254**, neutral tab **1255**, CP (for example, control pilot or control power) tab **1256**, phase 1 output (PH1) **1261**, and phase 2 output (PH2) 1262.
[0182] As depicted, power module **1200** has a distance of 44.5 mm by 31 mm. As depicted, the distance between tabs **1251** and **1252** is 4.5 mm, as is the distance between tabs **1253** and **1254**, as is the distance between tabs **1254** and **1256**. The distance between tab **1261** and an adjacent control pin is 3.6 mm. Other distances are possible.
[0183] FIG. **13** depicts an electrical schematic of an electrical inverter **1300**, according to one or more embodiments. Electrical inverter **1300** is a two-phase inverter, having HV1+, HV1−, HV2+, and HV2− inputs; and PH1 and PH2 outputs.
[0184] FIG. **14** depicts an exemplary layout for a three-phase inverter package **1400**, according to one or more embodiments. Inverter package **1400** may include power modules **1420**, **1422**, and **1424** positioned over a cooling rail **1470**, cooled by fluid **1480**. Examples of power modules **1420**, **1422**, and **1424** include, but are not limited to those depicted in FIGS. **8**, **11**, and **12**.
[0185] FIG. **15** depicts an electrical schematic of an electrical inverter **1500**, according to one or more embodiments. Electrical inverter **1500** is an Active Neutral Point Clamping (ANPC) three level inverter. Electrical inverter **1500** may be integrated and packaged into the electrical packages discussed herein, for instance in power module **1800** discussed with respect to FIG. **18**.
[0186] FIG. **16** depicts an electrical schematic of an electrical inverter **1600**, according to one or more embodiments. Electrical inverter **1600** is a T-type three-level inverter. Electrical inverter **1600** may be integrated and packaged into the electrical packages discussed herein, for instance in power module **1800** discussed with respect to FIG. **19**.
[0187] FIG. **17** depicts an exemplary three-phase inverter package **1700** of an inverter circuit, according to one or more embodiments. Package **1700** may include power modules **1720**, **1722**, and **1724**, which are APNC inverters.
[0188] Inverter package **1700** may include power modules **1720**, **1722**, and **1724** positioned over a cooling rail **1770**, cooled by fluid **1780**. Examples of power modules **1720**, **1722**, and **1724** include, but are not limited to those depicted in FIGS. **8**, **11**, and **12**.
[0189] Case **1730** may include a positive voltage input, a negative voltage input, and/or a neutral output. These connections are passed across one or more plates on case **1730**. The plates connect to the power modules via various terminals **1704**. For example, a first plate may conduct power from the positive voltage input to the power modules. A second plate may conduct power from negative the voltage input to the power modules. A third plate may conduct power from the power modules to the neutral output.
[0190] In turn, each of power modules **1720**, **1722**, and **1724** converts DC to AC power and outputs

AC power via one or more phase terminals **1705**. Phase terminals **1705** are assembled onto busbar assembly **1760**. Case **1730** may include one or more capacitors. Each capacitor may correspond to one of power modules **1720**, **1722**, and **1724**.

[0191] FIG. **18** depicts an exemplary architecture for a power module **1800**, according to one or more embodiments. Power module **1800** is an APNC inverter. Power module **1800** may include various connections or tabs, such as positive voltage (HV+)**1851**, negative voltage (HV−) **1852**, neutral (N) **1855**, phase 1 output (PH1) **1861**, and phase 2 output (PH2) **1862**.

[0192] As can be seen, power module **1800** has dimensions 59 mm×31 mm, but other dimensions are possible.

[0193] FIG. **19** depicts an exemplary architecture for a power module, according to one or more embodiments. Power module **1900** is a T-type inverter.

[0194] Power module **1900** may include various connections or tabs, such as positive voltage (HV+) **1951**, negative voltage (HV−) **1952**, neutral (N) **1955**, and phase 2 output (PH2) **1962**. As depicted, power module **1900** has dimensions 40 mm×31 mm, but other dimensions are possible.

[0195] FIG. **20** depicts an assembly **2000** of a power module with devices on a single substrate, according to one or more embodiments. Assembly **2000** may include top substrate **2010** and bottom substrate **2012**.

[0196] In some devices, having components placed on both a top and a bottom of a substrate, during assembly, the two substrates must be joined, face-to-face, such that necessary electrical connections between the components on the top substrate and bottom substrate are made. For example, connections between gates may be only 1 millimeter by 1 millimeter, requiring precise alignment of the top substrate and bottom substrate. Improper registration during assembly may lead to production loss and higher costs. Therefore, the power module assembled via assembly **2000** may provide all components placed on a single (e.g., exactly one, only one, or no more than one) substrate, such as top substrate **2010**, for example, thus avoiding issues of improper registration during assembly when the top substrate **2010** is joined with bottom substrate **2012**.

[0197] FIG. **21** depicts an assembly **2100** of a power module with devices on a single substrate, according to one or more embodiments. Assembly **2100** is a resulting assembly of the two layers depicted in FIG. **20**.

[0198] FIG. **22** depicts an assembly of a power module with devices on a single substrate, according to one or more embodiments. Physical construction **2200** may include stages **2201**, **2202**, **2203**, **2204**, **2205**, and **2206**, but other stages are possible. Each stage represents the addition of one or more components to the assembly.

[0199] Physical construction **2200** may further include source substrate **2211**, source-side solder connections **2212**, various components **2213**, drain side solder connections **2214**, and drain substrate **2215**. Examples of components **2213** include, but are not limited to spacers, die, and thermistors. In an example, 10 spacers, 12 dies, and 2 thermistors are included.

[0200] At stage **2201**, source substrate **2211** is integrated with two layers of polymide with embedded metal. At stage **2202**, the integration created at stage **2201**, e.g., substrate **2211** and polymide, is integrated with solder connections **2212**. At stage **2203**, the integration created at stage **2202** is integrated with components **2213**. At stage **2204**, the integration created at stage **2203** is integrated with the drain side solder connection **2214**. At stage **2205**, the integration created at stage **2204** is integrated with the drain substrate **2215**, creating the final package.

[0201] FIG. **23** depicts an assembly **2300** of a power module with devices on a source substrate and a drain substrate, according to one or more embodiments.

[0202] Assembly **2300** may include source metal layer (inner) **2301**, bottom dielectric **2302**, drain metal **2303**, source solder **2304**, source substrate **2305**, second metal **2306**, drain substrate **2307**, dielectric **2308**, source metal (outer) **2309**, top dielectric **2310**, drain metal (inner) **2311**, and drain solder **2312**.

[0203] FIG. **24** depicts an exemplary architecture for a power module **2400** including a thermistor, according to one or more embodiments. As discussed below, power module **2400** may include various layers of metal, substrate, and other components.

[0204] Power module **2400** may include upper substrate **2405**U and lower substrate **2405**L, overlay adhesive **2404**, metal layers **2410**, lead frame **2460** and **2465**, SiC die **2415**, drain connection **2420**, gate connection **2425**, source connection **2430**, polyimide or flex circuit **2435**, and base adhesive **2436**.

[0205] The upper substrate **2405**U and lower substrate **2405**L may be made of ceramic, e.g., silicon nitride (Si.sub.3N.sub.4) middle layers having thick metallization, e.g., direct bond copper (DBC) or active metal brazing (AMB) that may be employed on an outer surface and an inner surface of the ceramic middle layer.

[0206] Each of the metallization layers **2410** may include multiple conductive regions electrically separated from each other. The power module **2400** may include a semiconductor (e.g., silicon carbide (SiC)) die **2415** that has a drain connection **2420** to the upper substrate **2405**U. The source connection **2430** may be attached to the lower substrate **2405**L.

[0207] A polyimide or flex circuit **2435** including a base polyimide material with an internal copper layer **2440** may be used, thus making interconnect possible to the gate connection **2425** of the SiC MOSFET **2415**. Other components of power module **2400** may include lead frame **2460** and **2465** for connection to the positive supply voltage and negative supply voltage. The assembly may be over-molded with a dielectric material **2402**. The module may include one or more electrically conductive spacers that provide high current interconnect between an inner surface of upper substrate **2405**U and an inner surface of lower substrate **2405**L.

[0208] An example of a thickness of the metal layers is 400 um+/−0.03 um. An example of a thickness of the substrate layers **2405**U/L is 320 um+/−0.05 um. An example of a thickness of the gate is 58 um. An example of a thickness of the source is 100 um+/−25 um. An example of a thickness of the lead frame is 640 um. An example of a thickness of the base polyimide is 12 um. An example of a thickness of the base adhesive is 12 um. But other examples are possible.

[0209] Power module **2400** may include overlay adhesive **2404**. In an example, the overlay adhesive is 25 um thick. Power module **2400** may include overlay polyimide above overlay adhesive **2404**. In an example, the overlay polyimide is 12 um.

[0210] One or more embodiments may provide technical advantages. For example, disclosed systems include switchable inverter circuits that facilitate reconfiguration of inverters to and from different winding configurations, integrated power modules that include such switchable inverters, and integrated packages that include multiple power modules with thermal management. While the examples discussed herein describe electrical inverters for use in driving electric motors and/or in battery charging circuits, the disclosure is not limited thereto. Rather, the disclosed inverters may be used in any circuit.

[0211] Other embodiments of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the invention being indicated by the following claims.

## Claims

**1**. A system comprising: an inverter to convert DC power from a voltage source to AC power to drive a winding of a motor, wherein the inverter includes: a first power module including: a first inverter to be connected to a first end of the winding and to be connected to the voltage source, wherein the first inverter is configured to output first AC power to the first end of the winding; a second inverter to be connected to a second end of the winding and to be connected to the voltage source, wherein the second inverter is configured to output second AC power to the second end of the winding; and a changeover switch to selectively connect and disconnect the second end of the winding to and from a neutral connection.

**2**. The system of claim 1, wherein the changeover switch is further configured to change a configuration of the winding from Delta to Wye or from Wye to Delta.

**3**. The system of claim 1, wherein the first inverter is a first half bridge inverter and the second inverter is a second half bridge inverter.

**4**. The system of claim 1, wherein: the first inverter includes one or more first upper phase switches and one or more first lower phase switches, and the second inverter includes one or more second upper phase switches and one or more second lower phase switches.

**5**. The system of claim 1, wherein: the first inverter includes three upper phase switches and three lower phase switches to drive a first winding of the motor, and the second inverter includes three upper phase switches and three lower phase switches to drive the first winding of the motor.

**6**. The system of claim 1, wherein the inverter further includes one or more controllers to control an operation of the first inverter, the second inverter, and the changeover switch.

**7**. The system of claim 1, wherein: the first power module is configured to drive a first winding of the motor, and the inverter further includes: a second power module to drive a second winding of the motor; and a third power module to drive a third winding of the motor.

**8**. The system of claim 1, wherein the inverter further includes: a capacitor.

**9**. The system of claim 1, further including: the voltage source configured to supply the DC power to the inverter; and the motor configured to receive the AC power from the inverter to drive the motor, wherein the system is provided as a vehicle including the inverter, the voltage source, and the motor.

**10**. A system comprising a power module for an inverter, the power module including: a first inverter to be connected to a first end of a winding of a motor and to be connected to a voltage source, wherein the first inverter is configured to output first AC power to the first end of the winding; a second inverter to be connected to a second end of the winding and to be connected to the voltage source, wherein the second inverter is configured to output second AC power to the second end of the winding; and a changeover switch to selectively connect and disconnect the second end of the winding to and from a neutral connection.

**11**. The system of claim 10, wherein the changeover switch is further configured to change a configuration of the winding from Delta to Wye or from Wye to Delta.

**12**. The system of claim 10, wherein the first inverter is a first half bridge inverter and the second inverter is a second half bridge inverter.

**13**. The system of claim 10, wherein: the first inverter includes three upper phase switches and three lower phase switches to drive the winding of the motor, and the second inverter includes three upper phase switches and three lower phase switches to drive the winding of the motor.

**14**. The system of claim 10, wherein the changeover switch includes four switches.

**15**. The system of claim 10, wherein the second inverter is configured to be disabled below a power threshold to reduce switching loss of the power module.

**16**. The system of claim 15, wherein the changeover switch is configured to provide a current return path from the winding of the motor to the neutral connection when the second inverter is disabled.

**17**. The system of claim 15, wherein the changeover switch is configured to provide a connection to the winding of the motor for boost voltage during charging of the voltage source.

**18**. A system comprising: an inverter to convert DC power from a voltage source to AC power to drive a motor, wherein the inverter includes: a first power module to drive a first winding of the motor; a second power module to drive a second winding of the motor; and a third power module to drive a third winding of the motor, wherein each of the first power module, the second power module, and the third power module includes: a first inverter to be connected to a first end of a respective winding of the motor and to be connected to a voltage source, wherein the first inverter is configured to output first AC power to the first end of the respective winding; a second inverter to be connected to a second end of the respective winding and to be connected to the voltage source, wherein the second inverter is configured to output second AC power to the second end of the respective winding; and a changeover switch to selectively connect and disconnect the second end of the respective winding to and from a neutral connection.

**19**. The system of claim 18, wherein: each first inverter is a first half bridge inverter including three upper phase switches and three lower phase switches, and each second inverter is a second half bridge inverter including three upper phase switches and three lower phase switches.

**20**. The system of claim 18, wherein the inverter is configured to operate in a first configuration with each second inverter is disabled, and a second configuration with each first inverter and each second inverter enabled.